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% AE 6353 - Homework 2
% Lauren Forcey
% 9/28/25

function [xf,Cf,Sf] = laguerre(x0,t0,t,alpha,r0vec,v0vec,n,err)
x = x0;          % input initial guess
diff = 100;      % large IC
i = 0;
mu_e = 3.986e5;
r0 = norm(r0vec);
v0 = norm(v0vec);

while diff > err
    i = i + 1;

    % calculate y then C and S using it
    y = alpha*x^2;
    [C, S] = calcCandS(y);

    % calculate sigma0
    sig0 = dot(r0vec,v0vec)/sqrt(mu_e);

    % initial calculation F
    F = sig0*x^2*C + (1-r0*alpha)*x^3*S + r0*x - sqrt(mu_e)*(t-t0);

    % take derivative dF/df
    Fp = sig0*x*(1-y*S)+(1-r0*alpha)*x^2*C+r0;

    % take derivative again
    Fpp = sig0*(1-y*C)+((1-r0*alpha)*(1-y*S-C));

    % dealing with the plus mins in the denom
    denomp = Fp + sqrt(abs((n-1)^2*(Fp^2)-n*(n-1)*F*Fpp));
    denommm = Fp - sqrt(abs((n-1)^2*(Fp^2)-n*(n-1)*F*Fpp));

    if abs(denomp) > abs(denommm)
        denom = denomp;
    else
        denom = denommm;
    end

    num = n*F;

    % x_i+1
    xil = x - num/denom;

    % calculate delta
    diff = abs(xil-x);
    x = xil;
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fprintf('Iter %d: x = %.10f,S = %.5e,C = %.5e,F = %.5e,delta = %.2e\n',i,x,S,C,F,diff);  
end
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% gotta return the x value when it converges  
xf = x;  
Cf = C;  
Sf = S;  
end
```